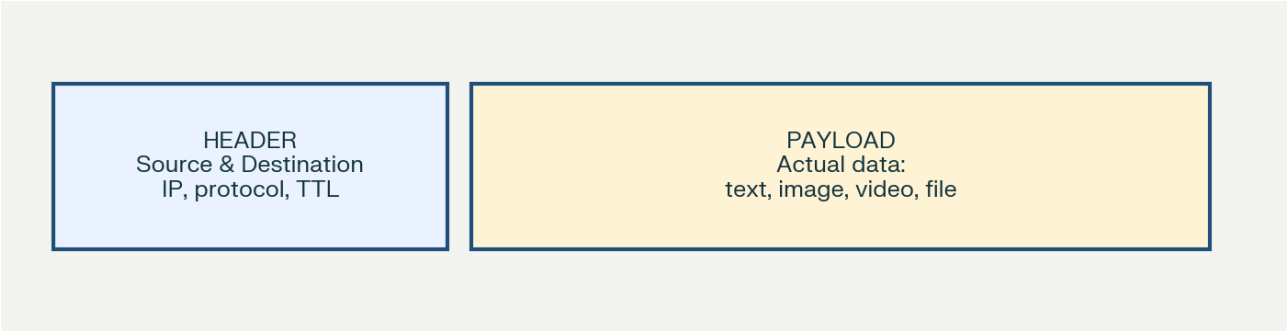
Unit 5: Network Protocols & Applications

Quick Revision Notes — Simple Language

1. Protocol and Packet

A **protocol** is a set of rules used for communication between devices. It tells what data is sent, how it is sent, and when it is sent.

- **Management protocols:** SNMP, ICMP — monitor and troubleshoot networks.
- **Communication protocols:** TCP, UDP, IP, HTTP, FTP, SMTP — transfer data.
- **Security protocols:** HTTPS, TLS, SSH — protect data.
- A **packet** is a small unit of data sent over a network.
- **Header:** source/destination IP, protocol, length, sequence number, TTL.
- **Payload:** actual data such as text, image, video, or file.



2. Internet Protocol (IP)

IP delivers packets from the source device to the destination device using IP addresses.

- IP works at the **Network Layer (Layer 3)** of the OSI model.
- IP is **connectionless**: it does not create a fixed connection before sending packets.
- IP works with TCP and UDP.
- IPv4 was created first; IPv6 was created because IPv4 addresses were running out.

3. IPv4

IPv4 uses a **32-bit** address written in four octets, such as **192.168.1.10**.

- IPv4 packet is called a **datagram**.
- Header minimum size: **20 bytes**; maximum with options: 60 bytes.
- **TTL:** maximum number of routers a packet can cross.
- **Header checksum:** checks errors in the header.
- **Source address:** sender IP; **Destination address:** receiver IP.

4. IPv6

IPv6 uses a **128-bit** address and provides a very large address space.

- Example: **2001:0db8:85a3::8a2e:0370:7334**
- Basic IPv6 header is fixed at **40 bytes**.
- IPv6 uses **Hop Limit** instead of TTL.
- IPv6 has **no header checksum**.
- IPv6 was mainly created to solve the shortage of IPv4 addresses.

IPv4 vs IPv6

Feature	IPv4	IPv6
Address size	32 bits	128 bits
Addresses	2^{32} (about 4.3 billion)	2^{12}■ (very large)
Example	192.168.1.10	2001:db8::1
Header size	20–60 bytes	Fixed 40 bytes
Header checksum	Present	Not present
Hop control	TTL	Hop Limit

5. IP Address Classes

Look at the first number of an IP address to find its class.

Class A	0–127	Large networks	255.0.0.0 /8
Class B	128–191	Medium networks	255.255.0.0 /16
Class C	192–223	Small networks	255.255.255.0 /24
Class D	224–239	Multicast	—
Class E	240–255	Experimental	—

6. Subnet Mask

A subnet mask separates an IP address into a **network part** and a **host part**. Network bits are 1 and host bits are 0.

- Class A: 255.0.0.0 = /8
- Class B: 255.255.0.0 = /16
- Class C: 255.255.255.0 = /24
- **Network address:** first address of a network.
- **Broadcast address:** last address of a network.
- **Usable hosts:** total addresses − 2.

7. Subnetting and CIDR

Subnetting divides one large network into smaller networks. CIDR writes an IP address with a prefix, such as 192.168.1.0/26.

- Number of subnets = 2^N, where N = borrowed host bits.
- Total addresses per subnet = 2^H, where H = remaining host bits.
- Usable hosts per subnet = $2^H - 2$.
- Example: 192.168.1.0/26 → borrowed bits = 26 – 24 = 2 → 4 subnets.
- Host bits = 32 – 26 = 6 → 64 total addresses → 62 usable hosts.
- Subnet mask for /26 = 255.255.255.192.

8. WWW and Web Hosting

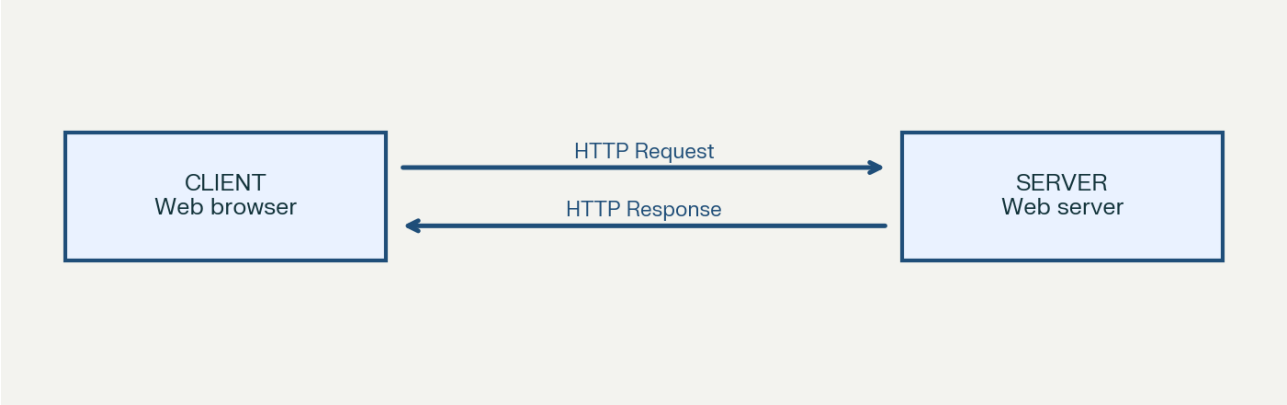
The World Wide Web is a collection of connected web pages and resources accessed through the Internet.

- Web pages are made using HTML and connected using hyperlinks.
- A URL gives the address of a web resource.
- **Web hosting** stores website files on a server and makes them available online.
- Hosting steps: register domain → choose host → upload files → set up database → configure DNS → publish → maintain.
- Types: Shared, VPS, Cloud, Dedicated, Reseller hosting.

9. HTTP

HTTP is used to send requests from a browser to a web server and receive web pages in response.

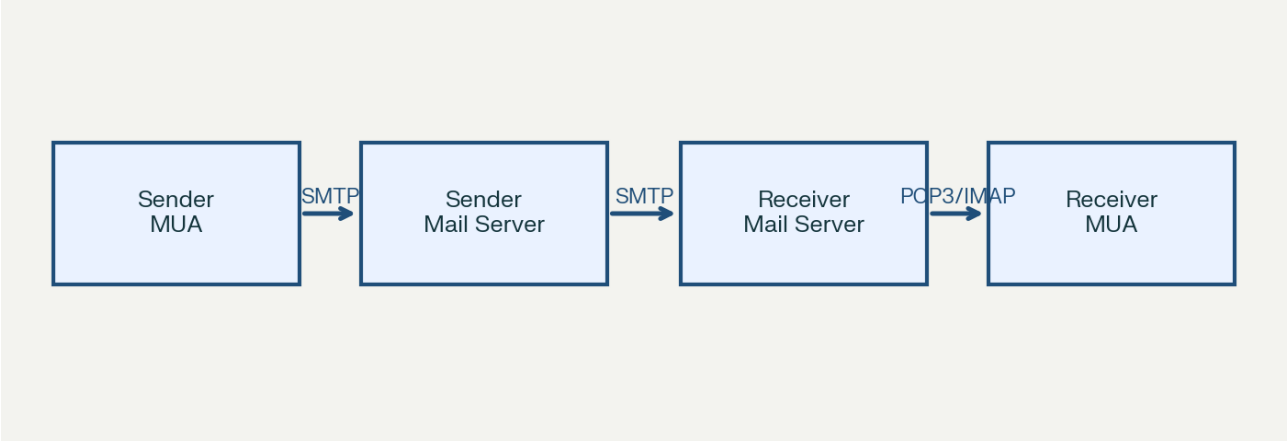
- HTTP commonly uses TCP port **80**.
- HTTP is **stateless**: the server does not remember previous requests automatically.
- **Non-persistent HTTP:** connection closes after one response.
- **Persistent HTTP:** connection stays open for multiple requests.



10. Email and SMTP

Email is used to send and receive messages, files, and attachments over the Internet.

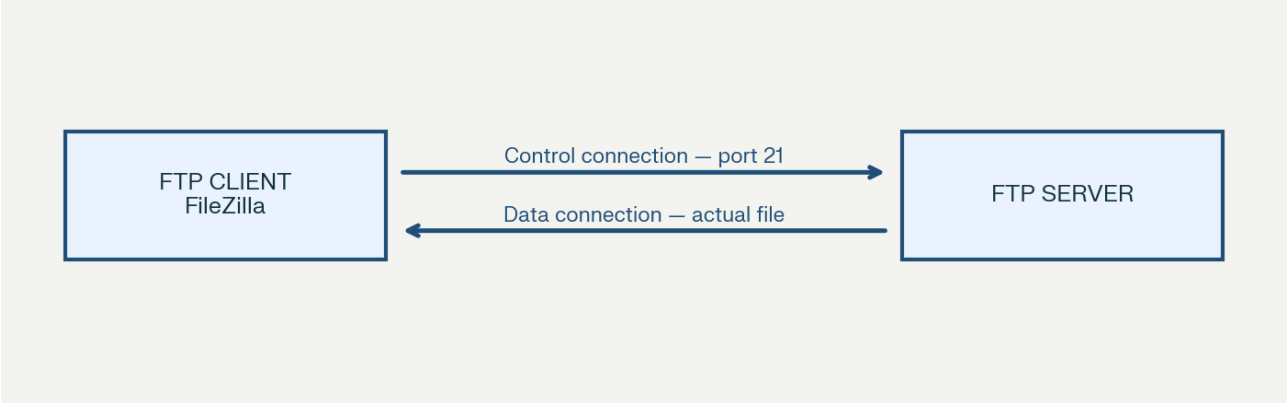
- Email address example: **inform@gmail.com** → username @ domain name.
- Email has a header, body, and attachments.
- **MUA:** Mail User Agent — Gmail, Outlook.
- **MTA:** Message Transfer Agent — transfers mail between servers.
- **SMTP** sends email; **POP3** and **IMAP** receive email.
- SMTP uses TCP port **25**.



11. FTP

FTP is used to transfer files between a client and a server.

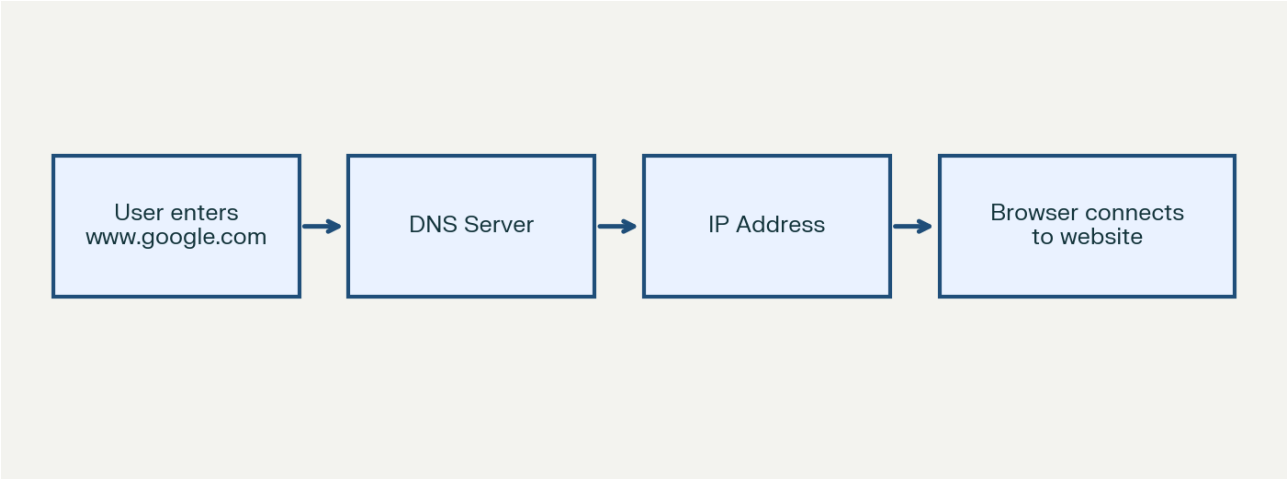
- FTP uses TCP and follows client-server architecture.
- **Control connection:** port 21; sends commands such as login and upload.
- **Data connection:** transfers the actual file.
- Example software: FileZilla.



12. DNS

DNS converts a domain name, such as www.google.com, into an IP address that computers can use.

- DNS is an application-layer protocol.
- DNS hierarchy: Root server → TLD server → Authoritative server.
- Root servers direct queries to TLD servers such as .com, .org, .edu.
- Authoritative servers give the final IP address of a domain.



13. Network Applications

Network applications are programs that use a network to provide services.

- Email, web browsing, remote login, file sharing, online games, video streaming, voice over IP, video conferencing, and social networking.

14. Last-Minute Memory Points

- IP delivers packets; HTTP delivers web pages.
- SMTP sends email; POP3/IMAP receive email.
- FTP transfers files; DNS converts names into IP addresses.
- IPv4 = 32 bits; IPv6 = 128 bits.
- Class A /8, Class B /16, Class C /24.
- Network address = first address; broadcast address = last address.
- Usable hosts = total addresses − 2.
- Subnets = 2^N; addresses per subnet = 2^H.